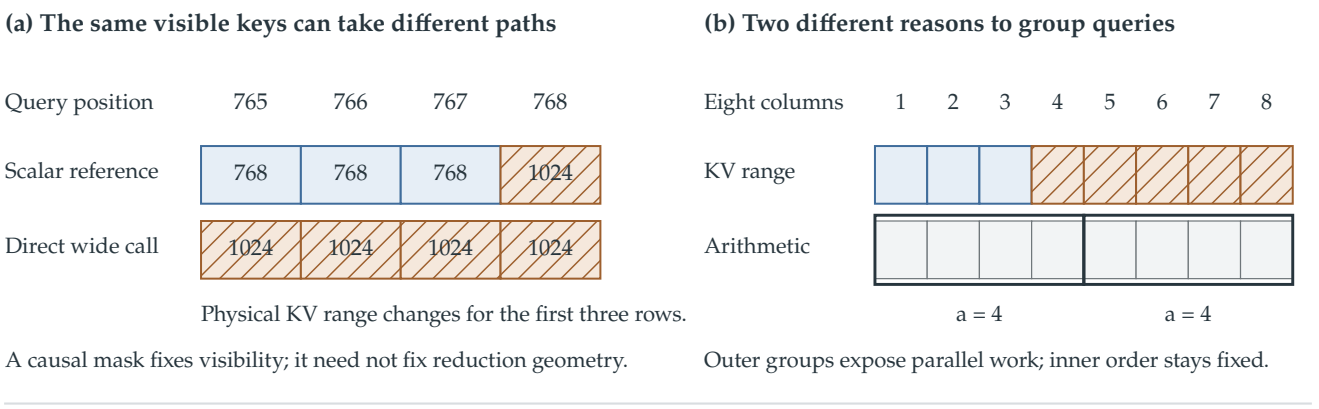

(a) The same visible keys can take different paths
Query position
765
766
767
768
Scalar reference
768
768
768
1024
Direct wide call
1024
1024
1024
1024
Physical KV range changes for the first three rows.
A causal mask fixes visibility; it need not fix reduction geometry.
(b) Two different reasons to group queries
Eight columns
1
2
3
4
5
6
7
8
KV range
Arithmetic
a = 4
a = 4
Outer groups expose parallel work; inner order stays fixed.
(c) Cut at the affected consumer, then rejoin in query order
Known queries
QKV projection wide group
Attention: 3 rows
Attention: 1 row
O / FFN projection wide group
Full vectors → next consumer
KV and pending state → reuse
768 / 1024 KV entries